

Class12 HW

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One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

```
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"
genotype <- read.table(url, header = TRUE)
head(genotype)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
nrow(genotype)
```

```
[1] 462
```

```
table(genotype$geno)
```

```
A/A A/G G/G
108 233 121
```

The sample size for each genotype is as follows: A/A=108, A/G=233. G/G=121

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
genotype %>%  
  group_by(geno) %>%  
  summarize(  
    sample_size = n(),  
    median_expression = median(exp)  
  )
```

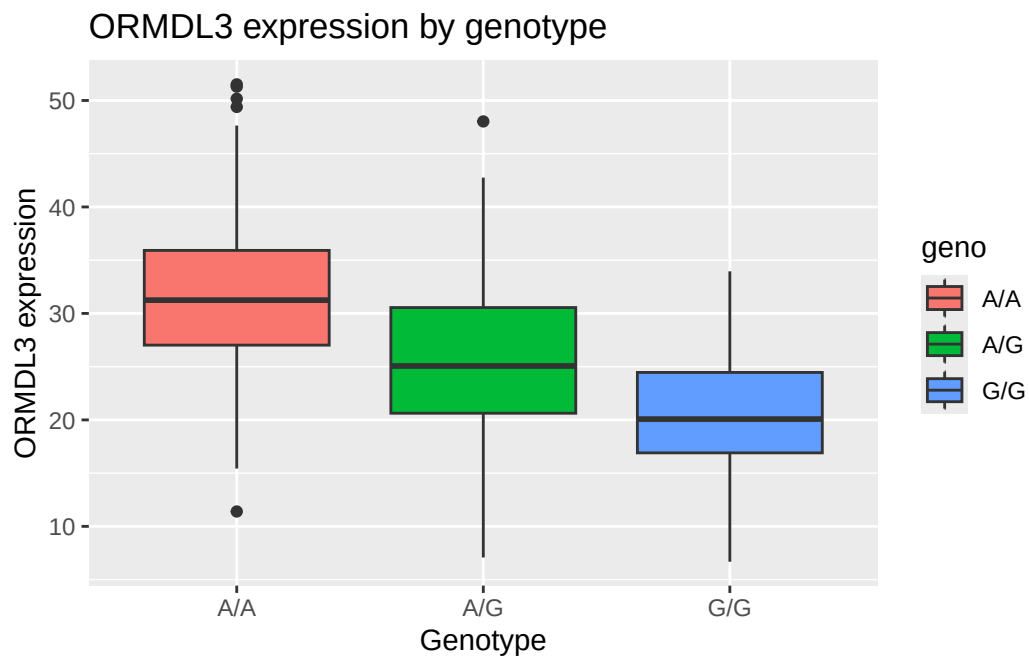
```
# A tibble: 3 x 3  
  geno sample_size median_expression  
  <chr>      <int>          <dbl>  
1 A/A         108           31.2  
2 A/G         233           25.1  
3 G/G         121           20.1
```

The median expression is A/A=31.25, A/G=25.06, G/G=21.07

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)  
  
ggplot(genotype, aes(x = geno, y = exp, fill = geno)) +  
  geom_boxplot() +  
  labs()
```

```
x = "Genotype",  
y = "ORMDL3 expression",  
title = "ORMDL3 expression by genotype"  
)
```



The boxplot shows that A/A has the highest ORMDL3 expression. This suggests that the SNP is associated with ORMDL3 expression. The G allele appears to lower expression.